

Question 1

Not yet answered

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Determine the domains of the functions.

 $f(x) = \sqrt{1-x}$ is defined on $(-\infty, \boxed{a}]$. $\boxed{a}$: $f(x) = \frac{1}{x^2-4}$ is defined on $(-\infty, \boxed{b}) \cup (\boxed{c}, \boxed{d}) \cup (\boxed{e}, \infty)$. $\boxed{b}$: $\boxed{c}$: $\boxed{d}$: $\boxed{e}$: $f(x) = \log(1-x^2)$ is defined on $(\boxed{f}, \boxed{g})$. $\boxed{f}$: $\boxed{g}$: **Question 2**

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Calculate $\lim_{x \rightarrow 2} \frac{x^2 + \sqrt{x + \sqrt{2x}}}{5x} = \frac{\boxed{a}}{\boxed{b}}$. $\boxed{a}$: $\boxed{b}$: Calculate $\lim_{x \rightarrow \infty} \frac{2x^3 + 2x + 4}{3x^3 - 5x^2 + x} = \frac{\boxed{c}}{\boxed{d}}$. $\boxed{c}$: $\boxed{d}$: **Question 3**

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Compute $\lim_{x \rightarrow 0} \frac{\sqrt{1-x} - \sqrt{1+x}}{x} = \boxed{a}$. $\boxed{a}$: Hint: use $\sqrt{a} - \sqrt{b} = \frac{(\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})}{(\sqrt{a} + \sqrt{b})}$.

Question 4

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Calculate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 - 1} = \frac{\boxed{a}}{\boxed{b}}$.

If we define $f(x) = \frac{x^2 - 1}{x^3 - 1}$ for $x \neq 1$ and $\frac{\boxed{c}}{\boxed{d}}$ for $x = 1$, $f(x)$ is continuous on $\mathbb{R}$.

Question 5

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Calculate $(\frac{3}{2})^4 = \frac{\boxed{a}}{\boxed{b}}$, $9^{-\frac{5}{2}} = \frac{\boxed{c}}{\boxed{d}}$, $\log_2 \sqrt{128} = \frac{\boxed{e}}{\boxed{f}}$, $\log_{\frac{1}{2}} \frac{1}{8} = \boxed{g}$

Question 6

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Calculate $\sin \frac{13\pi}{3} = \frac{\sqrt{\boxed{a}}}{\boxed{b}}$, $\cos(-\frac{3\pi}{4}) = \frac{\boxed{c}}{\sqrt{\boxed{d}}}$, $\sin \frac{\pi}{8} = \frac{\sqrt{\boxed{e}} - \sqrt{\boxed{f}}}{\boxed{g}}$.

Question 7

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Calculate $\lim_{n \rightarrow \infty} \frac{2^n + 1}{3^n} = \boxed{a}$, $\lim_{n \rightarrow \infty} \frac{\sqrt{4^n + 3}}{2^n} = \boxed{b}$, $\lim_{n \rightarrow \infty} (1 + 2^n)^{\frac{1}{n}} = \boxed{c}$, $\lim_{n \rightarrow \infty} (1 + \frac{1}{2^n})^n = \boxed{d}$.

Hint: Compare with a simpler limit and use squeezing.

Question 8

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Calculate $\lim_{x \rightarrow 0} \frac{x}{\sin(2x)} = \frac{\boxed{a}}{\boxed{b}}$, $\lim_{x \rightarrow 0} \frac{\log(x+1)}{\sin(x)} = \boxed{c}$, $\lim_{x \rightarrow \infty} x(e^{\frac{1}{x}} - 1) = \boxed{d}$.

Question 9

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Let $f(x) = x^2 - 4x + 5$ defined on $[0, 5]$. f takes the minimum at $x = \boxed{a}$ and its value $f(\boxed{a}) = \boxed{b}$. f takes the maximum at $x = \boxed{c}$ and its value $f(\boxed{c}) = \boxed{d}$.

$\boxed{a}$: $\boxed{b}$: $\boxed{c}$: $\boxed{d}$:

Question 10

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Let $f(x) = \sin(x^2)$, $g(x) = e^{-x^2}$, $h(x) = \log(x^2 + 1)$ on $(-\infty, \infty)$.

Choose the one which takes both maximum and minimum, and calculate these values.

Maximum: Minimum:

Choose all functions that are not bounded.

☐ f

☐ g

☐ h